EDUAI ACADEMIC SMART NOTES

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University Curriculum Series

# Full-Stack Web Engineering with React & FastAPI

## Section Topic: Backend Services with FastAPI & Pydantic

**Executive Summary:** Foundational review of Backend Services with FastAPI & Pydantic in full-stack web applications.

## 1. Core Learning Objectives

- Asynchronous route handlers
- Data contract validation
- Dependency injection mechanics

## 2. Conceptual Breakdown & Technical Formulations

Comprehensive engineering principles and code patterns for Backend Services with FastAPI & Pydantic.

### 1. Technical Framework

Detailed exploration of core patterns.

### 2. Practical Code

```
// Sample implementation
const [data, setData] = useState([]);
```

## 3. Self-Assessment & Exam Checklist

| Topic Milestone | Mastery Check |
|---|---|
| Can you formulate the primary mathematical loss / model objective? | [ ] Confirmed |
| Can you distinguish parametric vs non-parametric assumptions? | [ ] Confirmed |
| Are you prepared to calculate precision, recall, and F1 metrics? | [ ] Confirmed |
| Do you understand the difference between L1 sparsity and L2 shrinkage? | [ ] Confirmed |